Discrete Mathematics

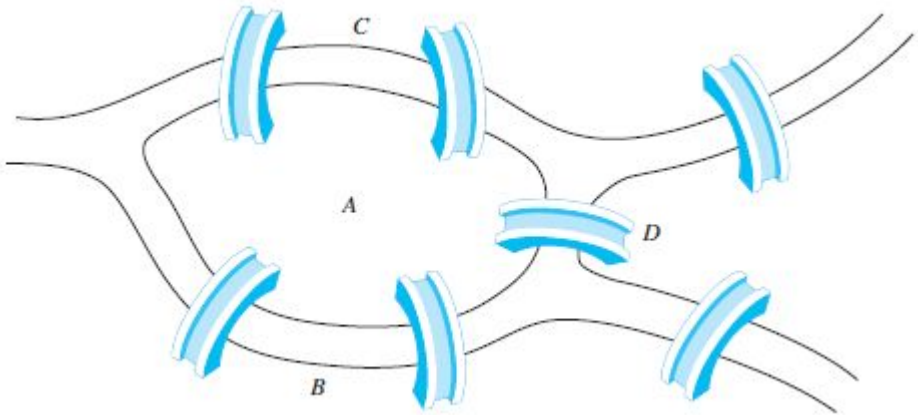
Week 14: Graphs Part 3

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History of Graph

The town of Königsberg, Prussia (now called Kaliningrad and part of the Russian republic), was divided into 4 sections by the branches of the Pregel River.

In the 18th century, 7 bridges of Königsberg connected these regions.

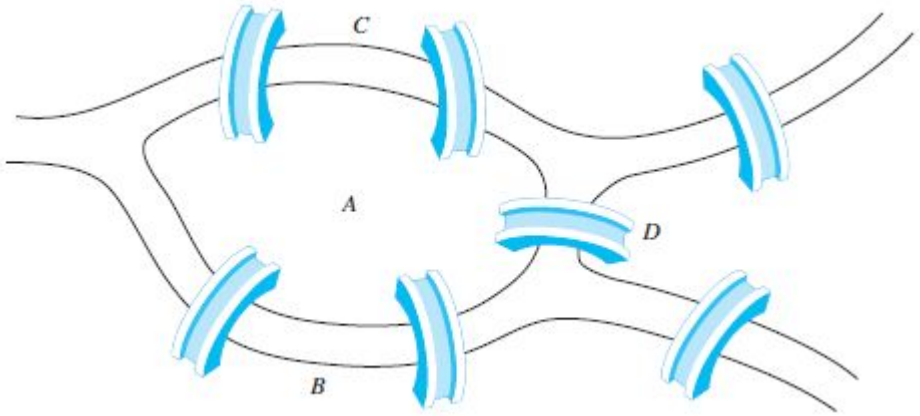


History of Graph



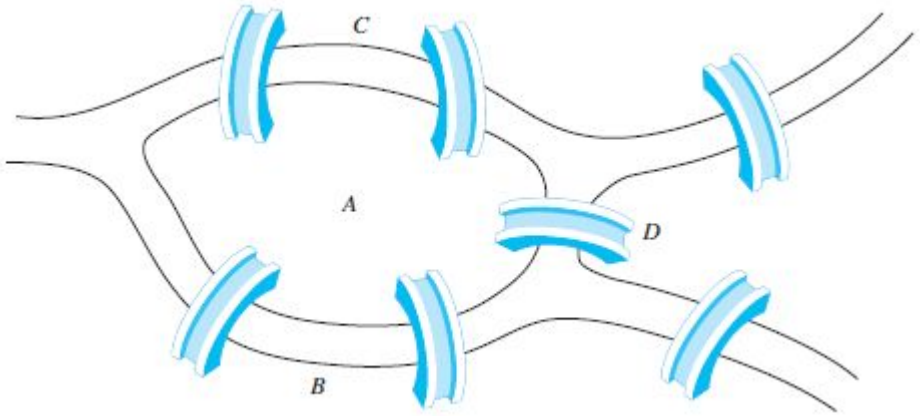
History of Graph

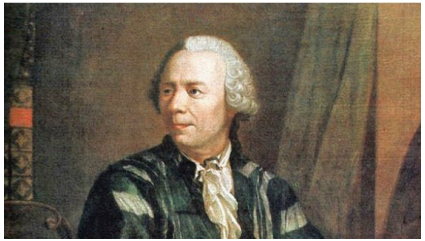
The townspeople wondered whether it was possible to start at some location in the town, travel across all the bridges once without crossing any bridge twice, and return to the starting point.



Seven Bridges of Königsberg

In 1736, the Swiss mathematician Leonhard Euler proved that ....





5. Euler and Hamiltonian Graphs

Can we travel along the edges of a graph starting at a vertex and returning to it by traversing each edge of the graph exactly once?

- Does a graph has an ***Euler circuit***?
- Can be easily answered by examining the degrees of vertices of the graph.

Similarly, can we travel along the edges of a graph starting at a vertex and returning to it while visiting each vertex of the graph exactly once?

- Does a graph has an ***Hamilton circuit***?
- Difficult to solve for most graphs

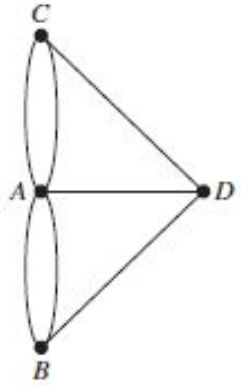
Euler Circuits and Paths

Definition 1:

An ***Euler circuit*** in a graph *G* is a simple circuit containing every edge of *G*.

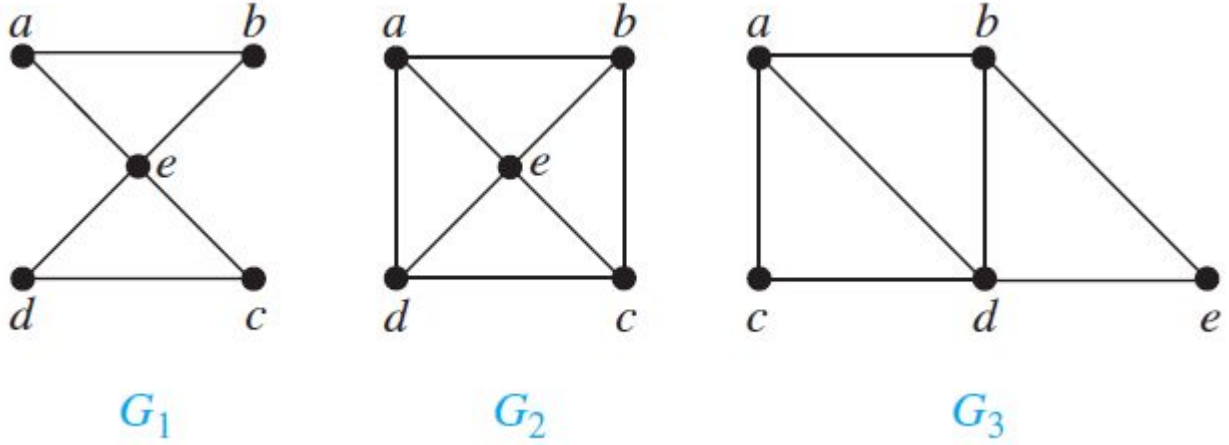
An ***Euler path*** in *G* is a simple path containing every edge of *G*.

- Recall that a path or circuit is simple if contains each edge exactly once.



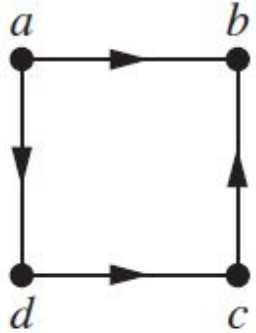
Euler Circuits and Paths Example

Example 1: Which of the undirected graphs have an Euler circuit? Of those that do not, which have an Euler path?

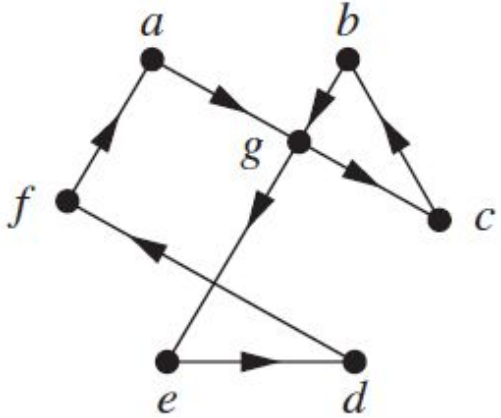


Euler Circuits and Paths Example

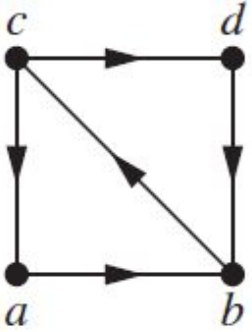
Example 2: Which of the directed graphs have an Euler circuit? Of those that do not, which have an Euler path?



H_1



H_2



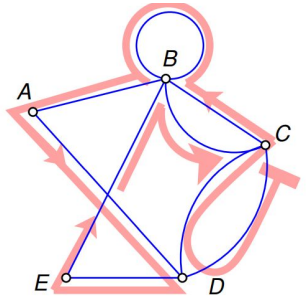
H_3

Necessary Conditions for Euler Circuits

Simple criteria for determining whether a multigraph has an Euler circuit or path.

- Necessary condition: Every vertex must have even degree.

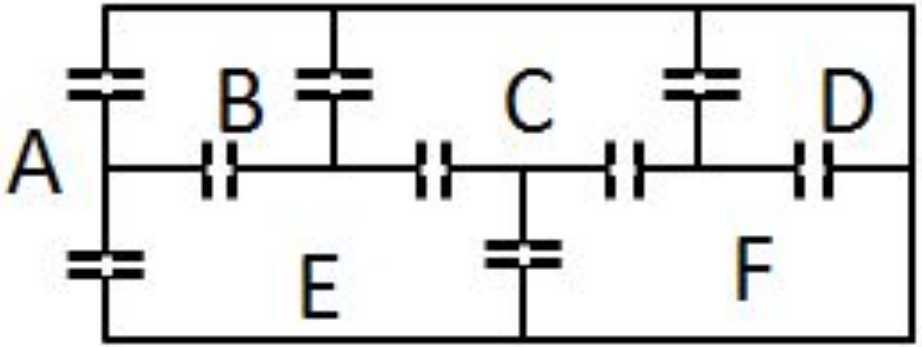
Theorem 1: A connected multigraph with ≥ 2 vertices has an Euler circuit if and only if each of its vertices has an even degree.



CDCBBADEBC

Necessary Conditions for Euler Circuits

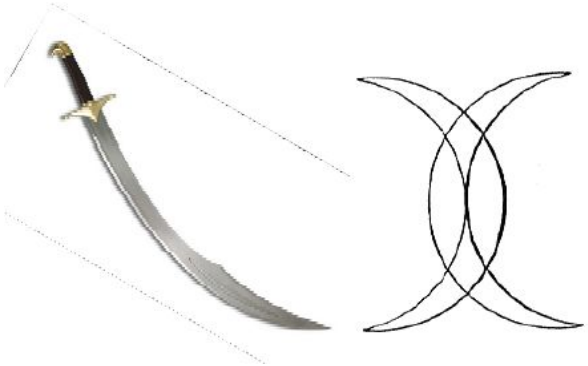
Example 3: Given a directory plan of a building, is it possible to enter the building from A and exit to room A by going through every door only once and every room must be visited?



Mohammed’s Scimitars Problem

Can you draw a picture in a continuous motion without lifting a pencil?

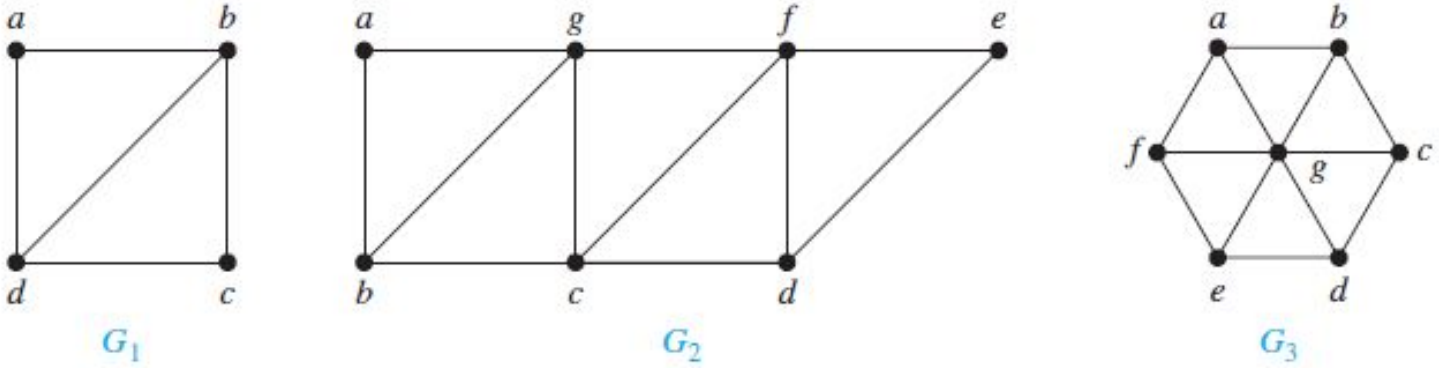
For example, can Mohammed’s scimitars be drawn in this way, where the drawing begins and ends at the same point?



Necessary Conditions for Euler Paths

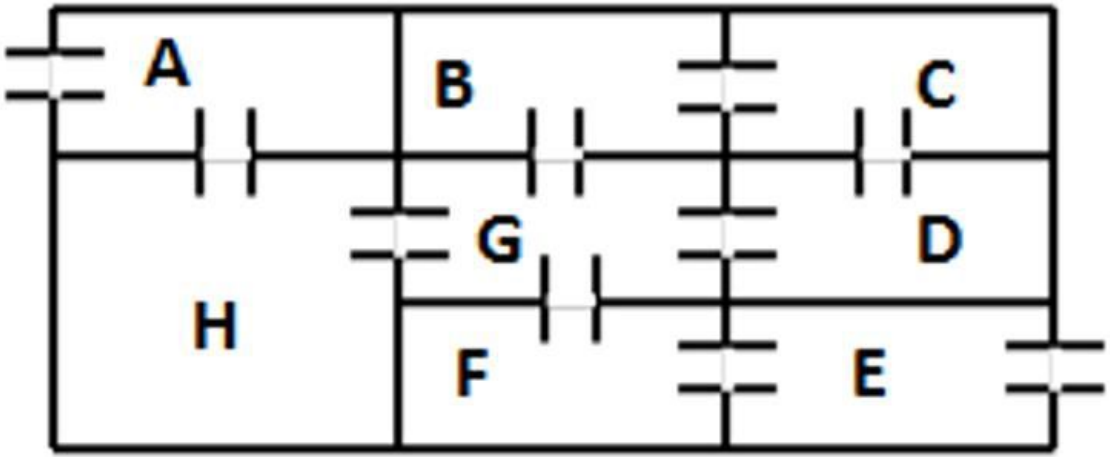
Theorem 2: A connected multigraph has an Euler path but not an Euler circuit if and only if it has exactly 2 vertices of odd degree.

Example: Which graphs have an Euler path?



Necessary Conditions for Euler Paths

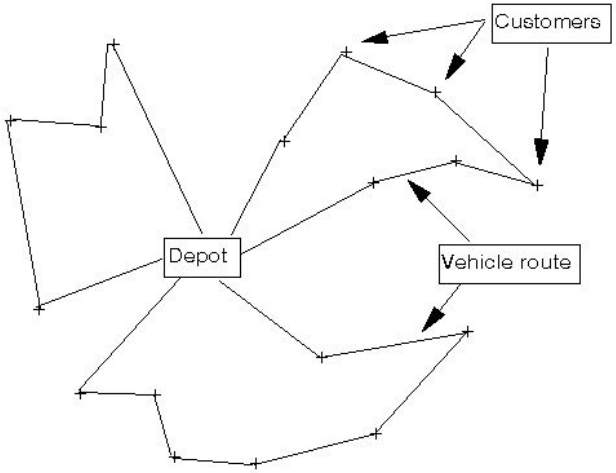
Example 4: Given a directory plan of a building, is it possible to enter the building to room A and exit from room E by going through every door only once and every room must be visited?



Applications of Euler Paths and Circuits

Euler paths and circuits can be used to solve many practical problems such as finding a path or circuit that traverses each

- street in a neighborhood,
- road in a transportation network,
- connection in a utility grid,
- link in a communications network, exactly once.



Other applications are found in the

- layout of circuits,
- network multicasting,
- molecular biology, where Euler paths are used in the sequencing of DNA.

Hamiltonian Circuits and Paths



Definition 2:

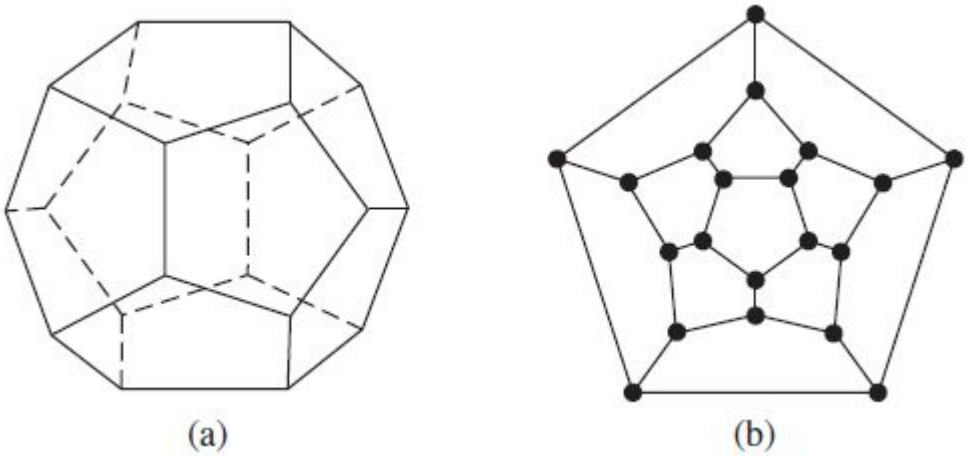
A simple circuit in a graph *G* that passes through every vertex exactly once is called a ***Hamilton circuit***.

A simple path in a graph *G* that passes through every vertex exactly once is called a ***Hamilton path*.**

Hamilton’s Puzzle - A Voyage Round the World

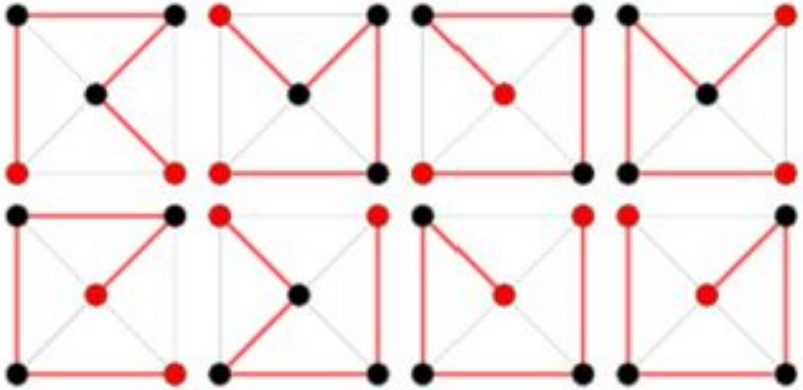
Hamilton terminology comes from a game, called the **Icosian puzzle**, invented in 1857 by the Irish mathematician Sir William Rowan Hamilton.

It consisted of a wooden dodecahedron [a polyhedron with 12 regular pentagons as faces], with a peg at each vertex of the dodecahedron, and string.

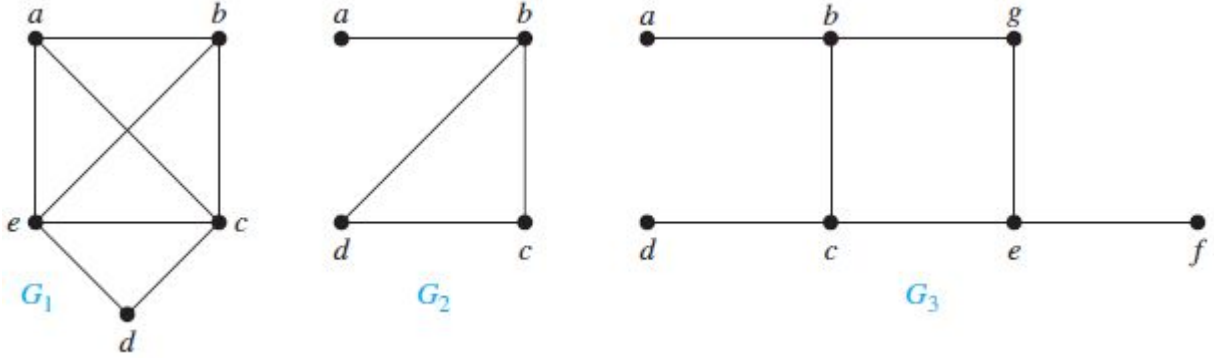


Hamilton Circuits and Paths Examples





Hamilton Circuits and Paths Exercise



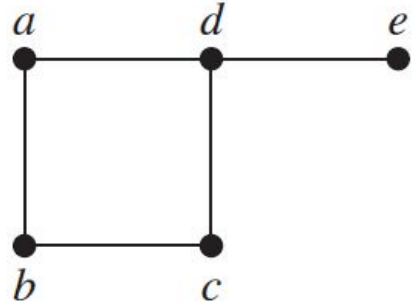
Exercise 1: Which of the simple graphs have a Hamilton circuit or, if not, a Hamilton path?

Conditions for the Existence of Hamilton Circuits

Unlike for an Euler circuit, no simple necessary and sufficient conditions are known for the existence of a Hamilton circuit.

Only a few sufficient conditions have been found.

- The more edges for each vertex, the more likely to have a Hamilton circuit.



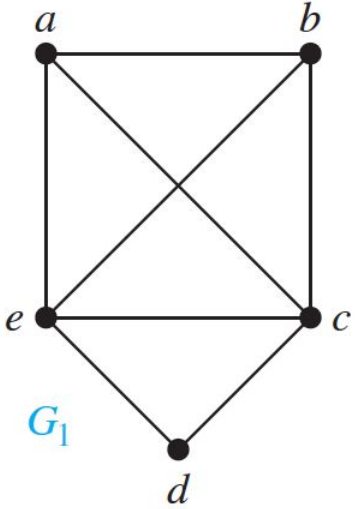
Conditions for the Existence of Hamilton Circuits

Dirac's Theorem: If G is a simple graph with $n \geq 3$ vertices such that the degree of every vertex in G is $\geq n/2$, then G has a Hamilton circuit.

Ore's Theorem: If G is a simple graph with $n \geq 3$ vertices such that $\deg(u) + \deg(v) \geq n$ for every pair of nonadjacent vertices, then G has a Hamilton circuit.

Hamilton Circuits Example

Example 1: Show that a graph has a Hamilton circuit.



n =

deg(a) =

deg(b) =

deg(c) =

deg(d) =

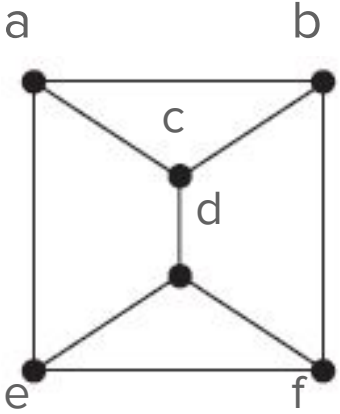
deg(e) =

Dirac's Theorem: n ≥ 3 vertices and each node has degree ≥ n/2

Ore's Theorem: n ≥ 3 vertices such that deg(u) + deg(v) ≥ n for

Hamilton Circuits Exercise

Exercise: Show that a graph has a Hamilton circuit.



n =

deg(a) =

deg(b) =

deg(c) =

deg(d) =

deg(e) =

deg(f) =

Dirac's Theorem: n ≥ 3 vertices and each node has degree ≥ n/2

Ore's Theorem: n ≥ 3 vertices such that deg(u) + deg(v) ≥ n for

Applications of Hamilton Paths and Circuits

Hamilton paths and circuits can be used to solve many practical problems such as finding a path or circuit that visits each

- road intersection,
- place pipelines in a utility grid,
- node in a communications network, exactly once.

The famous **traveling salesman problem** asks for the shortest route a traveling salesperson should take to visit a set of cities.

- A solution involves finding a Hamilton circuit with smallest total weight of edges.

Shortest-Path Problems

Graphs that have a number assigned to each edge are called **weighted graphs**.

- What is a **shortest path**, that is, a path of least length, between two given vertices?

